# Cirrus SR20 G3 / G6 (SR20)

Continental IO-360-ES, 200 hp (G3) / Lycoming IO-390-C3B6, 215 hp (G6) · 4 (G6: 5) seats · Typical SR20 G3 (IO-360-ES, Avidyne or Perspective) and G6 (IO-390, Perspective+) POH procedures; speeds are G3 figures, G6 differs by a few knots.

## V-speeds & limits

| | |
|---|---|
| Vr | **65–70 KIAS (flaps 50%)** |
| Vx | **81 KIAS** |
| Vy | **96 KIAS** |
| Best glide | **87–88 KIAS** |
| Vfe | **119 KIAS (50%) / 104 KIAS (100%)** |
| Va | **130 KIAS (3,050 lb)** |
| Vno | **163 KIAS** |
| Vne | **200 KIAS** |
| Vpd (CAPS max demonstrated) | **133 KIAS** |
| Vs / Vso | **65 / 56 KIAS** |
| Vref | **75–80 KIAS (flaps 100%)** |

## Normal procedures

### Before Start

- Preflight inspection — **COMPLETE (fuel drains, oil 6–8 qt, pitot cover removed)**
- CAPS handle safety pin — **REMOVED (handle cover closed, pin stowed)**
- Passenger briefing — **COMPLETE (belts, doors, CAPS location & use, exits)**
- Seats, belts, harnesses — **ADJUST & LOCK**
- Doors — **LATCHED (both handles)**
- Brakes — **TEST & SET (parking brake)**
- Fuel selector — **FULLER TANK (LEFT or RIGHT)**
- Power lever — **IDLE**
- Mixture — **CUTOFF**
- Alternate induction air — **CLOSED (pushed in)**
- Bat 2 — **ON – check volts 23–25 V**
- Bat 1 — **ON**
- Avionics — **OFF (Perspective: avionics master OFF)**
- Annunciators / CAS — **CHECK (no red)**
- Fuel quantity — **CHECK (vs visual check)**
- Flaps — **CHECK operation (50%, 100%), set UP**
- Strobes / nav lights — **ON**

## Engine Start

| | |
|---|---|
| Propeller area | **CLEAR** |
| Power lever | **FULL FORWARD** |
| Mixture | **FULL RICH** |
| Fuel pump | **PRIME (hold 2–3 s cold, none when hot), then BOOST** |
| Power lever | **OPEN 1/4 in** |
| Ignition | **START (max 20 s, release when engine fires)** |
| Power lever | **1,000 RPM** |
| Oil pressure | **CHECK within 30 s (green 30–60 psi)** |
| Alt 1 / Alt 2 | **ON – check charging, no ALT 1 / ALT 2 caution** |
| Fuel pump | **BOOST (leave on until after takeoff)** |
| Avionics | **ON** |
| Engine instruments | **CHECK** |
| Pitot heat | **TEST then OFF** |

## Before Taxi

| | |
|---|---|
| Flight instruments / PFD | **CHECK & SET (altimeter, HSI, CDI)** |
| Radios / transponder | **SET** |
| Trim | **SET for takeoff (neutral mark)** |
| Flaps | **50%** |
| Parking brake | **RELEASE** |
| Brakes | **CHECK immediately after moving** |

Nosewheel is free-castoring: steer with differential braking at low speed. Avoid prolonged brake drag – brake overheat is a fire risk.

| | |
|---|---|
| HSI / turn indication | **CHECK during turns** |

## Before Takeoff

| | |
|---|---|
| Parking brake | **SET** |
| Doors | **LATCHED** |
| CAPS handle | **VERIFY pin removed, cover closed** |
| Seat belts / harnesses | **SECURE** |
| Flight controls | **FREE & CORRECT** |
| Fuel selector | **FULLER TANK** |
| Fuel pump | **BOOST** |
| Mixture | **FULL RICH (lean for field elevation above 3,000 ft)** |
| Power lever | **1,700 RPM** |
| Alt 1 / Alt 2 | **CHECK positive amps, Ess Bus / Main Bus volts green** |
| Magnetos | **CHECK L / R (max drop 150 RPM, max diff 75 RPM)** |
| Engine instruments | **CHECK green (oil temp min 100°F for takeoff)** |
| Power lever | **IDLE check (600–800 RPM), then 1,000 RPM** |
| Flaps | **50% (check indication)** |
| Trim | **SET takeoff** |
| Autopilot | **CHECK / disconnect** |
| Pitot heat / lights | **AS REQUIRED** |
| Transponder | **ALT** |
| Takeoff briefing | **COMPLETE (Vr 65–70, Vy 96, abort point, CAPS available above approx 600 ft AGL)** |

### After Takeoff / Climb

| | |
|---|---|
| Airspeed | **Vy 96 KIAS (obstacles: Vx 81 KIAS)** |
| Flaps | **UP (above 80 KIAS, positive rate)** |
| Fuel pump | **OFF above 1,000 ft AGL (BOOST if fuel flow fluctuates)** |
| Power | **FULL (cruise climb 110–120 KIAS)** |
| Mixture | **FULL RICH (lean per fuel flow placard above 3,000 ft)** |
| Engine instruments | **CHECK (CHT max 460°F, oil temp max 240°F)** |

### Cruise

| | |
|---|---|
| Power | **SET (typ. 55–75%, 2,500–2,700 RPM)** |
| Mixture | **LEAN (per power table / lean-assist, 50°F rich of peak or LOP as approved)** |
| Fuel pump | **OFF (BOOST if fuel flow fluctuates or hot/high)** |
| Fuel selector | **SWITCH tanks every 30 min, keep within 10 gal imbalance** |
| Fuel quantity / totalizer | **CHECK vs flight plan** |
| Engine instruments | **CHECK** |

### Descent

| | |
|---|---|
| Altimeter | **SET** |
| Mixture | **ENRICH as altitude decreases** |
| Power | **REDUCE gradually (keep CHT above 250°F)** |
| Fuel selector | **FULLER TANK** |
| Approach briefing | **COMPLETE (ATIS, approach, speeds)** |

### Before Landing

| | |
|---|---|
| Seats / belts | **SECURE** |
| Fuel selector | **FULLER TANK** |
| Fuel pump | **BOOST** |
| Mixture | **FULL RICH (as required for altitude)** |
| Flaps | **50% below 119 KIAS, 100% below 104 KIAS** |
| Airspeed | **FINAL 75–80 KIAS (100%), 80–85 KIAS (50%)** |
| Autopilot | **OFF** |
| Landing light | **ON** |

### After Landing

| | |
|---|---|
| Flaps | **UP** |
| Fuel pump | **OFF** |
| Pitot heat | **OFF** |
| Transponder | **STBY / as required** |
| Lights | **AS REQUIRED** |
| Power lever | **1,000 RPM taxi** |

### Shutdown

Parking brake ..... **SET**

Power lever ..... **IDLE, 1 min cool-down**

Avionics ..... **OFF**

Mixture ..... **CUTOFF**

Ignition ..... **OFF (after prop stops), key removed**

Alt 1 / Alt 2 ..... **OFF**

Bat 1 / Bat 2 ..... **OFF**

CAPS handle safety pin ..... **INSERT**

Fuel selector ..... **LEFT or RIGHT (leave on a tank)**

Pitot cover / chocks / tie-downs ..... **INSTALL**

## Abnormal procedures

### ALT 1 or ALT 2 Failure

ALT 1 / ALT 2 caution, Main or Ess Bus amps zero

ALT 1 / ALT 2 circuit breaker (affected) ..... **CHECK IN**

Alt 1 / Alt 2 (affected) ..... **OFF, then ON (reset once)**

**▶ Alternator restored?**

**Yes:**

Electrical load ..... **MONITOR**

**No – ALT 1 failed:**

Alt 1 ..... **OFF**

Non-essential loads ..... **OFF (Alt 2 supplies approx 20 A to the essential bus only)**

Bat 1 ..... **MONITOR – powers main bus until depleted (approx 30 min)**

Deferred: ALT 1 failed – main bus on battery, limited endurance; flaps once on final, land as soon as practical. ..... **NOTED**

**No – ALT 2 failed:**

Alt 2 ..... **OFF**

Flight ..... **CONTINUE – Alt 1 supplies both buses**

Deferred: ALT 2 failed – no electrical redundancy, avoid IMC / night if practical, land as soon as practical. ..... **NOTED**

### Low Volts / Both Alternators Failed

LOW VOLTS warning on Main and Ess bus

Alt 1 / Alt 2 ..... **OFF then ON (reset once)**

**▶ Voltage recovered?**

**Yes:**

Electrical load ..... **MONITOR**

**No:**

Alt 1 / Alt 2 ..... **OFF**

Bat 1 ..... **OFF (save Bat 2 for essential bus)**

Non-essential loads ..... **OFF**

Bat 2 powers the essential bus (PFD, standby attitude, COM1, NAV1) for approx 30 min. Flaps and trim are on the main bus – turn Bat 1 ON briefly for flaps on final.

Deferred: Both alternators failed – battery only, flaps once on final with Bat 1 ON, land as soon as possible. ..... **NOTED**

## Low Oil Pressure / High Oil Temperature

OIL PRESS warning (below 30 psi, 10 psi idle), oil temperature above 240°F or CHT above 460°F

Oil pressure / temperature / CHT ..... **CHECK all, cross-check**

▶ **Which indication?**

**Low pressure:**

▶ **Oil temperature rising or pressure below 10 psi?**

**Yes:**

Power ..... **REDUCE to minimum required**

Landing site ..... **SELECT, expect engine failure; CAPS if no safe site**

Deferred: Low oil pressure – land as soon as possible, engine failure possible. ..... **NOTED**

**No – temperature normal:**

Flight ..... **CONTINUE, monitor closely**

Deferred: Suspect oil pressure indication – land as soon as practical. ..... **NOTED**

**High temperature:**

Mixture ..... **ENRICH**

Airspeed ..... **INCREASE (lower nose)**

Power ..... **REDUCE**

▶ **Temperature returning to green?**

**Yes:**

Flight ..... **CONTINUE, monitor**

**No:**

Power ..... **MINIMUM practical**

Deferred: Engine overheating – land as soon as practical, expect possible power loss. ..... **NOTED**

## Rough Running / Loss of Power

Fuel pump ..... **BOOST**

Mixture ..... **ADJUST (richer, then leaner)**

Fuel selector ..... **SWITCH tanks**

Alternate induction air ..... **OPEN (induction icing or blockage)**

Ignition ..... **CHECK L / R individually, BOTH**

▶ **Engine running smoothly?**

**Yes:**

Configuration ..... **KEEP the working combination**

Deferred: Engine ran rough – land as soon as practical, maintenance check. ..... **NOTED**

**No:**

Power ..... **REDUCE to smoothest setting**

Land ..... **AS SOON AS POSSIBLE (prepare for engine failure / CAPS)**

## Low Fuel Flow / Fuel Flow Fluctuation

Fuel flow fluctuating, FUEL PRESS or low fuel caution

Fuel pump ..... **BOOST**

Fuel selector ..... **SWITCH to fuller tank**

Mixture ..... **FULL RICH**

Fuel quantity ..... **CHECK both tanks**

▶ **Fuel flow steady?**

**Yes:**

Fuel pump ..... **LEAVE ON BOOST**

Deferred: Fuel pump on BOOST continuously – engine-driven pump or vapour suspect. ..... **NOTED**

**No:**

Land ..... **AS SOON AS POSSIBLE, expect engine failure**

## Flap Failure

FLAPS circuit breaker ..... **CHECK / RESET once**

Flap switch ..... **CYCLE (UP / 50% / 100%)**

### ▶ Flaps operable?

**Yes:**

Flap setting ..... **AS REQUIRED**

**No:**

Approach ..... **FLAPS UP 85–90 KIAS / 50% 80–85 KIAS, flat approach**

Deferred: Flaps inoperative – flaps-up landing, increased approach speed and landing distance. ..... **NOTED**

## Door Open in Flight

Airspeed ..... **REDUCE below 119 KIAS**

Door ..... **DO NOT attempt to close – hold shut if practical**

A door open in flight is noisy and causes buffet; the aircraft remains controllable.

Loose objects / belts ..... **SECURE**

Deferred: Door open – land as soon as practical, keep speed below 119 KIAS. ..... **NOTED**

## PFD / AHRS / ADC Failure

Display backup (red button) ..... **PRESS – reversionary mode on MFD**

Standby instruments ..... **USE (airspeed, attitude, altimeter, compass)**

Autopilot ..... **OFF if AHRS failed**

PFD / AHRS circuit breakers ..... **CHECK**

### ▶ Flight in IMC?

**Yes:**

ATC ..... **ADVISE, request radar assistance / nearest VMC**

Deferred: Primary flight display failed – partial panel, request vectors, avoid IMC. ..... **NOTED**

**No:**

Deferred: PFD / AHRS failed – standby instruments only. ..... **NOTED**

## Static Source Blocked / Erroneous Airspeed

Pitot heat ..... **ON**

Alternate static source ..... **OPEN (valve under panel, left of pilot)**

Airspeed / altitude ..... **CROSS-CHECK with GPS ground speed and standby altimeter**

Attitude / power ..... **FLY known pitch and power settings**

Deferred: Pitot-static unreliable – fly attitude and power, add margin to approach speed. ..... **NOTED**

# Emergency procedures

## CAPS Deployment

Loss of control, mid-air collision, structural failure, pilot incapacitation, or engine failure where a safe forced landing is not assured

| | |
|---|---|
| Airspeed (memory) | **MINIMISE (below 133 KIAS, as slow as practical)** |
| CAPS handle (memory) | **PULL straight down, both hands, full stroke (approx 45 lb)** |
| Mixture (memory) | **CUTOFF** |
| Fuel selector (memory) | **OFF** |
| Bat 1 / Bat 2 / Alt 1 / Alt 2 (memory) | **OFF** |
| Seat belts / harnesses (memory) | **TIGHT** |
| Brace (memory) | **ASSUME emergency landing body position (arms crossed, hands on shoulders)** |

CAPS is the primary response to loss of control, mid-air collision, structural failure, pilot incapacitation and engine failure over terrain where a safe forced landing is not assured. Recommended decision altitude 2,000 ft AGL; minimum demonstrated deployment from a loss of control 920 ft AGL, from level flight approx 400 ft AGL. Below the minimum, deploy anyway – it is still the best option.

| | |
|---|---|
| Ignition / fuel pump | **OFF (if time permits)** |
| ELT | **ON (if time permits)** |
| Descent | **EXPECT approx 1,700 ft/min, 8 s to full canopy, hard landing (approx 3 g)** |
| After touchdown | **EVACUATE upwind if canopy drags, aircraft may be dragged by wind** |

## Engine Failure During Takeoff Roll

| | |
|---|---|
| Power lever (memory) | **IDLE** |
| Brakes (memory) | **APPLY (differential for directional control)** |
| Flaps | **UP** |
| Mixture | **CUTOFF** |
| Fuel selector | **OFF** |
| Ignition | **OFF** |
| Bat 1 / Bat 2 / Alt 1 / Alt 2 | **OFF** |

## Engine Failure after Takeoff

Airborne, insufficient runway remaining

**LAND IMMEDIATELY**

| | |
|---|---|
| Airspeed (memory) | **BEST GLIDE or landing speed (80 KIAS flaps 50%)** |
| Mixture (memory) | **FULL RICH** |
| Fuel selector (memory) | **SWITCH tanks** |
| Fuel pump (memory) | **BOOST** |
| Alternate induction air (memory) | **OPEN** |
| Ignition (memory) | **BOTH (check)** |

Below approx 600 ft AGL land straight ahead with only small heading changes – do not turn back. Above approx 600 ft AGL CAPS deployment is an option if a safe landing area is not available.

### ▶ Safe landing area ahead?

**Yes – land ahead:**

| | |
|---|---|
| Flaps | **100% when landing assured** |
| Mixture | **CUTOFF** |
| Fuel selector | **OFF** |
| Ignition | **OFF** |
| Bat 1 / Bat 2 / Alt 1 / Alt 2 | **OFF** |
| Doors | **UNLATCH before touchdown** |

**No – CAPS:**

Deploy CAPS – see CAPS Deployment procedure.

## Engine Failure in Flight

Loss of power with altitude available

**LAND AS SOON AS POSSIBLE**

| | |
|---|---|
| Airspeed (memory) | **88 KIAS (best glide)** |
| Mixture (memory) | **FULL RICH** |
| Fuel selector (memory) | **SWITCH tanks** |
| Fuel pump (memory) | **BOOST** |
| Alternate induction air (memory) | **OPEN** |
| Ignition (memory) | **BOTH (cycle L / R / BOTH)** |
| Landing site | **SELECT, turn towards it** |

**▶ Restart attempt practicable?**

**Yes – restart:**

| | |
|---|---|
| Power lever | **1/2 in open** |
| Mixture | **FULL RICH, then lean slowly** |
| Ignition | **START if propeller stopped** |

**▶ Engine restarted?**

**Yes:**

| | |
|---|---|
| Power | **AS REQUIRED (minimum needed)** |
| Engine instruments | **MONITOR** |
| Deferred: Engine restarted after in-flight failure – land as soon as possible, suspect fuel or induction problem. | **NOTED** |

**No:**

Continue with forced landing or CAPS.

**No – forced landing / CAPS:**

**▶ Safe forced landing assured?**

**Yes – forced landing:**

| | |
|---|---|
| Mayday / 7700 | **TRANSMIT** |
| Seats / belts | **SECURE** |
| Flaps | **100% when landing assured** |
| Airspeed | **75–80 KIAS on final** |
| Mixture | **CUTOFF** |
| Fuel selector | **OFF** |
| Ignition | **OFF** |
| Bat 1 / Bat 2 / Alt 1 / Alt 2 | **OFF** |
| Doors | **UNLATCH before touchdown** |

**No – deploy CAPS:**

Deploy CAPS before descending through 2,000 ft AGL – see CAPS Deployment procedure.

## Engine Fire in Flight

**LAND IMMEDIATELY**

- Mixture (memory) ..... **CUTOFF**
- Fuel pump (memory) ..... **OFF**
- Fuel selector (memory) ..... **OFF**
- Power lever (memory) ..... **IDLE**
- Ignition (memory) ..... **OFF**
- Cabin heat / defrost (memory) ..... **OFF**
- Airspeed ..... **INCREASE (sideslip to keep flames from cabin)**

**▶ Fire extinguished and landing area within gliding range?**

**Yes:**

- Forced landing ..... **EXECUTE (do not attempt restart)**

**No:**

Deploy CAPS – see CAPS Deployment procedure.

## Engine Fire During Start

- Mixture (memory) ..... **CUTOFF**
- Fuel pump (memory) ..... **OFF**
- Fuel selector (memory) ..... **OFF**
- Power lever (memory) ..... **FULL FORWARD**
- Ignition (memory) ..... **CONTINUE cranking (draw fire into engine)**
- Ignition / Bat / Alt ..... **OFF when fire out**
- Fire extinguisher ..... **OBTAIN**
- Aircraft ..... **EVACUATE**

## Cabin / Electrical Fire

**LAND AS SOON AS POSSIBLE**

- Bat 1 / Bat 2 (memory) ..... **OFF**
- Alt 1 / Alt 2 (memory) ..... **OFF**
- Avionics (memory) ..... **OFF**
- Cabin heat / fresh air ..... **CLOSE**
- Fire extinguisher ..... **ACTIVATE**

After discharge, open fresh air vents / crack a door to ventilate once fire is out.

**▶ Fire out and electrical power required?**

**Yes:**

- Circuit breakers ..... **CHECK – do not reset tripped breakers**
- Bat 2 / Alt 2 ..... **ON (essential bus first)**
- Bat 1 / Alt 1 / avionics ..... **ON one at a time, check for smoke**
- Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable. ..... **NOTED**

**No:**

Continue with electrics off. Standby instruments and magnetic compass only; no flaps, no electric trim.

- Deferred: All electrics off – flaps-up landing approx 90 KIAS, no radios, no trim. ..... **NOTED**

## Emergency Descent

- Power lever (memory) ..... **IDLE**
- Mixture (memory) ..... **AS REQUIRED**
- Flaps (memory) ..... **UP**
- Airspeed (memory) ..... **163 KIAS smooth air / 130 KIAS turbulence**

## Inadvertent Spin Entry

CAPS handle (memory) ..... **PULL immediately**

The SR20 is not approved for spins and no spin recovery has been demonstrated – CAPS deployment is the only approved recovery. Minimum demonstrated deployment 920 ft AGL.

Power lever ..... **IDLE**

Controls ..... **NEUTRAL**

After deployment ..... **CONTINUE CAPS Deployment procedure**

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Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist dce92715 · 2026-11-07